

Ignacio Magaña Hernandez

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Employment

Carnegie Mellon University <i>McWilliams Postdoctoral Fellow</i>	Pittsburgh, PA <i>June 2023 - Present</i>
University of Wisconsin-Milwaukee <i>NSF Graduate Research Fellow</i>	Milwaukee, WI <i>September, 2017 - May 2023</i>
The Chinese University of Hong Kong <i>Undergraduate Research Assistant</i>	Hong Kong, Hong Kong <i>March, 2016 - January 2017</i>
Caltech <i>LIGO SURF Fellow</i>	Pasadena, CA <i>June, 2015 - September, 2015</i>

Education

University of Wisconsin-Milwaukee <i>PhD in Physics</i>	Milwaukee, WI <i>September 2017 - May 2023</i>
University of California, Santa Barbara <i>B.S., Physics (Highest Honors)</i>	Santa Barbara, CA <i>September 2014 - September 2017</i>
Foothill College <i>Transfer Student, Physics</i>	Los Altos Hills, CA <i>September 2012 - June 2014</i>

Awards & Fellowships

- **CMU McWilliams Postdoctoral Fellowship** 2023 - Present
McWilliams Center for Cosmology and Astrophysics.
- **CMU McWilliams Seed Grant** 2025
Internal grant funding exploratory projects to generate preliminary results for high impact research.
- **UWM Advanced Opportunity Program Fellowship** 2020 - 2023
UWM fellowship supporting underrepresented graduate students for three years.
- **NSF Graduate Research Fellowship** 2017 - 2022
NSF fellowship supporting graduate students for three years.
- **NASA Wisconsin Space Grant Consortium Fellowship** Summer 2020, 2021, 2022
NASA funded fellowship for graduate research in STEM at Wisconsin institutions.
- **UWM Physics Department Lichtman Fellowship** Spring 2020
Departmental fellowship awarded for exceptional academic achievement in physics.
- **UCSB Physics Department Highest Academic Honors** Summer 2017
- **LIGO SURF Blanco Fellowship** Summer 2015
Summer Research Fellowship at Caltech/LIGO, supporting undergraduates in GW science.
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- **NSF Science, Math and Engineering (S-STEM) Scholarship** Fall 2013 - Spring 2014
NSF scholarship providing support to students pursuing STEM degrees.

Publication Summary

h-index: 78 / 21 **Publications:** 140 / 44 **Citations:** 64k / 2.5k

First number includes LIGO collaboration papers; second number excludes them
Citations according to inspirehep.net in October 2026

First/Second Author Publications

1. *Normalizing Flow Emulator for Gravitational Wave Detection Probability in the LIGO/Virgo/KAGRA O4 Observing Run*
J. Gassert, **I. Magaña Hernandez**, A. J. Amsellem, A. Palmese and B. Póczos
([arXiv:2610.00306](https://arxiv.org/abs/2610.00306), 0 citations)

2. *Dark siren cosmology with the Dark Energy Spectroscopic Instrument*
I. Magaña Hernandez, A. Palmese et al.
 Under DESI internal review
3. *Data-Driven Trends and Subpopulations in the Gravitational Wave Binary Black Hole Merger Population with UMAP*
 A. J. Amsellem, **I. Magaña Hernandez**, A. Palmese and J. Gassert
 ([arXiv:2603.06566](#), [1 citations](#))
4. *Lucky Strikes: On the Origins of GW190814 Through Isolated Binary Evolution*
I. Magaña Hernandez and K. Breivik
 ([arXiv:2511.16648](#), [1 citations](#)) Submitted to *ApJL*
5. *Spectral siren cosmology from gravitational-wave observations in GWTC-4.0*
I. Magaña Hernandez and A. Palmese
 ([arXiv:2509.03607](#), [21 citations](#)) Submitted to *Physical Review Letters*
6. *Astrophysics informed Gaussian processes for gravitational-wave populations: Evidence for the onset of the pair-instability supernova mass gap*
I. Magaña Hernandez and A. Palmese
 ([arXiv:2508.19208](#), [16 citations](#)) Submitted to *Physical Review Letters*
7. *Evidence of a new feature in the binary black hole mass distribution at $70 M_{\odot}$ from gravitational-wave observations*
I. Magaña Hernandez and A. Palmese
 Phys. Rev. D 111, 083031 (2025) ([arXiv:2407.02460](#), [21 citations](#))
8. *Searching for binary black hole sub-populations in gravitational wave data using binned Gaussian processes*
 A. Ray, **I. Magaña Hernandez**, K. Breivik, J.D.E. Creighton
 ApJ 991 17 (2025) ([arXiv:2404.03166](#), [52 citations](#))
9. *Beyond Gaps and Bumps: Spectral Siren Cosmology with Non-Parametric Population Models*
I. Magaña Hernandez and A. Ray
 ([arXiv:2404.02522](#), [34 citations](#)) Submitted to *Physical Review D*
10. *On the association of GW190425 with its potential electromagnetic counterpart FRB 20190425A*
I. Magaña Hernandez, V. D’Emilio, S. Morisaki, M. Bhardwaj and A. Palmese
 ApJL 971 L5 (2024) ([arXiv:2404.02496](#), [14 citations](#))
11. *Non-parametric inference of the population of compact binaries from gravitational wave observations using binned Gaussian processes*
 A Ray, **I. Magaña Hernandez**, S. Mohite, J.D.E. Creighton and S. Kapadia
 ApJ 957 37 (2023) ([arXiv:2304.08046](#), [92 citations](#))
12. *Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers*
I. Magaña Hernandez
 ([arXiv:2211.01272](#), [11 citations](#)) Submitted to *Physical Review D*
13. *Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers*
I. Magaña Hernandez
 Phys. Rev. D 107, 084033 (2023) ([arXiv:2112.07650](#), [23 citations](#))
14. *A Bayesian statistical framework for identifying strongly-lensed gravitational-wave signals*
 R.K.L. Lo and **I. Magaña Hernandez**
 Phys. Rev. D 107, 123015 (2023) ([arXiv:2104.09339](#), [102 citations](#))
15. *Identifying strong gravitational-wave lensing during the second observing run of Advanced LIGO and Advanced Virgo*
 X. Liu, **I. Magaña Hernandez** and J.D.E. Creighton
 ApJ 908 97 (2021) ([arXiv:2009.06539](#), [108 citations](#))
16. *Cosmological inference using gravitational wave standard sirens: A mock data analysis*
 R. Gray, **I. Magaña Hernandez** et al.
 Phys. Rev. D 101, 122001 (2020) ([arXiv:1908.06050](#), [274 citations](#))

Other Short Author Work

1. *One Year and One Night to 1% in H_0 : Efficient Spectroscopic Strategy for Dark Siren Cosmology*
 Y. Dang, A. J. Amsellem, **I. Magaña Hernandez**, A. Palmese and B. S. Sathyaprakash
 ([arXiv:2607.17284](#), [0 citations](#))
2. *Electromagnetic Follow-up of the Sub-Solar Mass Gravitational Wave Candidate S251112cm: Kilonova Constraints and a Coincident IIb Supernova*
 X. J. Hall et al. (including **I. Magaña Hernandez**)

- ([arXiv:2605.10940](#), 8 citations)
3. *Wide Jets or Low Rates: Reconciling Short GRB and Gravitational-Wave Neutron Star Merger Rates*
K. Kunnumkai, A. Palmese, B. O'Connor, A. Farah and **I. Magaña Hernandez**
([arXiv:2604.05046](#), 6 citations)
 4. *Probing the environment around GW170817 with DESI: insights on galaxy group peculiar velocities for standard siren measurements*
A. J. Amsellem et al. (including **I. Magaña Hernandez**)
ApJ 1001, 157 (2026) ([arXiv:2512.08818](#), 5 citations)
 5. *AT2025ulz and S250818k: Investigating early time observations of a subsolar mass gravitational-wave binary neutron star merger candidate*
X. Hall et al. (including **I. Magaña Hernandez**)
([arXiv:2510.24620](#), 18 citations)
 6. *AT2025ulz and S250818k: Deep X-ray and radio limits on off-axis afterglow emission and prospects for future discovery*
B. O'Connor et al. (including **I. Magaña Hernandez**)
ApJL 995, L47 (2025) ([arXiv:2510.23728](#), 14 citations)
 7. *AT2025ulz and S250818k: Leveraging DESI spectroscopy in the hunt for a kilonova associated with a sub-solar mass gravitational wave candidate*
X. Hall et al. (including **I. Magaña Hernandez**)
ApJL 1001, L20 (2026) ([arXiv:2510.23723](#), 14 citations)
 8. *Kilonova Constraints for the LIGO/Virgo/KAGRA Neutron Star Merger Candidate S250206dm: GW-MMADS Observations*
L. Hu et al. (including **I. Magaña Hernandez**)
ApJL 990 L46 (2025) ([arXiv:2506.22626](#), 17 citations)
 9. *A Dark Siren Measurement of the Hubble Constant with the LIGO/Virgo Gravitational Wave Event GW190412 and DESI Galaxies*
W. Ballard, A. Palmese, **I. Magaña Hernandez** et al.
Res. Notes AAS 7 250 (2023) ([arXiv:2311.13062](#), 35 citations)
 10. *Follow-up analyses to the O3 LIGO–Virgo–KAGRA lensing searches*
J. Janquart et al. (including **I. Magaña Hernandez**)
MNRAS 526, 3832 (2023) ([arXiv:2306.03827](#), 83 citations)
 11. *Challenges for Fast Radio Bursts as Multi-Messenger Sources from Binary Neutron Star Mergers*
M. Bhardwaj, A. Palmese, **I. Magaña Hernandez**, V. D’Emilio and S. Morisaki
ApJ 977 122 (2024) ([arXiv:2306.00948](#), 31 citations)
 12. *Localization of binary neutron star mergers with a single Cosmic Explorer*
P. Baral, S. Morisaki, **I. Magaña Hernandez** and J.D.E. Creighton
Phys. Rev. D 108, 043010 (2023) ([arXiv:2304.09889](#), 17 citations)
 13. *The Hitchhiker’s guide to the galaxy catalog approach for gravitational wave cosmology*
J. Gair et al. (including **I. Magaña Hernandez**)
AJ 166 22 (2023) ([arXiv:2212.08694](#), 119 citations)
 14. *A joint ranking statistic for multi-messenger astronomical searches with gravitational waves*
B. Piotrkowski, A. Baylor and **I. Magaña Hernandez**
Class. Quantum Grav. 39 085010 (2022) ([arXiv:2111.12814](#), 8 citations)
 15. *Rapid model comparison of equations of state from gravitational wave observation of binary neutron star coalescences*
S. Ghosh, X. Liu, J.D.E. Creighton, **I. Magaña Hernandez** et al.
Phys. Rev. D 104, 083003 (2021) ([arXiv:2104.08681](#), 25 citations)
 16. *Gravitational wave friction in light of GW170817 and GW190521*
S. Mastroianni, L. Haegel, K. Karathanasis, **I. Magaña Hernandez** and D.A. Steer
JCAP 02 043 (2021) ([arXiv:2010.04047](#), 45 citations)
 17. *Current observations are insufficient to confidently associate the binary black hole merger GW190521 with AGN J124942.3 + 344929*
G. Ashton, K. Ackley, **I. Magaña Hernandez** et al.
Class. Quantum Grav. 38 235004 (2021) ([arXiv:2009.12346](#), 114 citations)
 18. *First measurement of the Hubble parameter from bright binary black hole GW190521*
S. Mukherjee et al. (including **I. Magaña Hernandez**)
([arXiv:2009.14199](#), 73 citations)
 19. *Bayesian inference for compact binary coalescences with BILBY: Validation and application to the first LIGO–Virgo*

gravitational-wave transient catalogue

I.M. Romero-Shaw et al. (including **I. Magaña Hernandez**)

MNRAS 499, 3295 (2020) ([arXiv:2006.00714](#), 642 citations)

20. *A statistical standard siren measurement of the Hubble Constant from the LIGO/Virgo gravitational wave compact object merger GW190814 and Dark Energy Survey galaxies*

A. Palmese et al. (including **I. Magaña Hernandez**)

ApJL 900 L33 (2020) ([arXiv:2006.14961](#), 174 citations)

21. *A standard siren measurement of the Hubble Constant from GW170817 without the electromagnetic counterpart*

M. Fishbach, R. Gray, **I. Magaña Hernandez** et al.

ApJL 871 L13 (2019) ([arXiv:1807.05667](#), 304 citations)

LIGO-Virgo Publications to which I highly contributed:

1. *GW190814: Gravitational Waves from the Coalescence of a 23 Solar Mass Black Hole with a 2.6 Solar Mass Compact Object*

LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)

ApJL 896 L44 (2020) ([arXiv:2006.12611](#), 2121 citations) **Led and wrote the Hubble constant analysis**

2. *A Gravitational-wave Measurement of the Hubble Constant Following the Second Observing Run of Advanced LIGO and Virgo*

LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)

ApJ 909 218 (2021) ([arXiv:1908.06060](#), 424 citations) **Paper writing and analysis team member**

3. *First measurement of the Hubble constant from a dark standard siren using the Dark Energy Survey galaxies and the LIGO/Virgo binary-black-hole merger GW170814*

M. Soares-Santos, A. Palmese et al. and LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)

ApJL 876 L7 (2019) ([arXiv:1901.01540](#), 377 citations) **Validated analysis using gwcsmo**

Invited Seminars

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|--|----------------|
| 1. American Museum of Natural History
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | October 2025 |
| 2. Vanderbilt University
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | April 2025 |
| 3. KAGRA Seminar
<i>Binary black hole populations and cosmology</i> | April 2025 |
| 4. Stanford University (KIPAC Seminar)
<i>Binary black hole populations and cosmology with gravitational-wave observations and DESI</i> | March 2025 |
| 5. Carnegie Mellon University (STAMPS Seminar)
<i>Binary black hole populations and cosmology</i> | January 2025 |
| 6. Montclair State University
<i>Cosmology, Lensing, and Modified Gravity with Gravitational Waves</i> | September 2023 |
| 7. Carnegie Mellon University
<i>Testing general relativity with strongly lensed gravitational waves</i> | September 2023 |
| 8. Cardiff University
<i>Cosmology and modified gravity with binary black holes</i> | September 2022 |
| 9. UC Berkeley
<i>Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers</i> | June 2022 |
| 10. University of Wisconsin-Milwaukee
<i>Cosmology and modified gravity with binary black hole mergers</i> | April 2022 |
| 11. Eotvos Lorand University
<i>Gravitational wave cosmology with large galaxy surveys</i> | September 2019 |
| 12. University of Glasgow
<i>Measuring the Hubble constant with gravitational-wave standard sirens</i> | August 2019 |
| 13. The Chinese University of Hong Kong
<i>A measurement of the Hubble constant from the second observation run of Advanced LIGO-Virgo</i> | February 2019 |
| 14. Nagoya University
<i>A measurement of the Hubble constant from the second observation run of Advanced LIGO-Virgo</i> | February 2019 |

Talks at Conferences

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|---|----------------|
| 1. Gravitational Waves Physics and Astronomy Workshop
<i>Dark Siren Cosmology with the Dark Energy Spectroscopic Instrument (DESI)</i> | December 2025 |
| 2. COSMO25
<i>Gravitational-wave dark siren cosmology with DESI</i> | October 2025 |
| 3. Multimessenger Astronomy in the Era of Foundational AI
<i>Trends in the GWTC-3 Binary Black Hole Merger Population with UMAP</i> | August 2025 |
| 4. Summer DESI Collaboration Meeting
<i>Gravitational-wave cosmology with DESI</i> | July 2025 |
| 5. APS Global Physics Summit
<i>Lucky Strikes: Unraveling the asymmetric origins of GW190814 through binary field evolution</i> | April 2025 |
| 6. The 6th neighborhood workshop
<i>Beyond Gaps and Bumps: Spectral Siren Cosmology with Non-Parametric Population Models</i> | April 2024 |
| 7. Gravitational-wave populations: what's next?
<i>Non-parametric inference of the population of compact binaries from gravitational wave observations using binned Gaussian processes</i> | July 2023 |
| 8. APS April Meeting
<i>Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers</i> | April 2023 |
| 9. 11th Australasian Conference on General Relativity and Gravitation
<i>Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers</i> | February 2022 |
| 10. Midwest Relativity Meeting
<i>Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers</i> | October 2021 |
| 11. Midwest Relativity Meeting
<i>On the candidate electromagnetic counterpart of GW190521 and its implications on the Hubble constant</i> | October 2020 |
| 12. Gravitational Waves Physics and Astronomy Workshop
<i>A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo</i> | October 2019 |
| 13. Multi-Messenger Astrophysics in the Gravitational Wave Era
<i>A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo</i> | September 2019 |
| 14. YITP Asian-Pacific Winter School and Workshop on Gravitation and Cosmology
<i>A standard siren measurement of the Hubble constant from GW170817 without the electromagnetic counterpart</i> | February 2019 |
| 15. Midwest Relativity Meeting
<i>Cosmological inference using gravitational wave standard sirens: A mock data challenge</i> | October 2018 |
| 16. 19th Annual Physical Society of Hong Kong
<i>Feedforward Seismic Noise Cancellation at the 40m Prototype Interferometer</i> | June 2016 |

Other Workshops

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| 1. Multi-messenger Transients from Binary Mergers and Stellar Explosions
<i>Aspen Center for Physics</i> | August 2024 |
| 2. Black Holes and Neutron Stars with Gravitational Waves
<i>Yukawa Institute for Theoretical Physics</i> | October 2019 |
| 3. The New Era of Gravitational-Wave Physics and Astrophysics
<i>Kavli Institute for Theoretical Physics</i> | May - July 2019 |
| 4. Asian-Pacific Winter School and Workshop on Gravitation and Cosmology
<i>Yukawa Institute for Theoretical Physics</i> | February 2019 |
| 5. Les Houches Summer School: Gravitational-Wave Astronomy
<i>Ecole de Physique des Houches</i> | July 2018 |
| 6. Summer School on Gravitational-Wave Astronomy
<i>ICTS-TIFR</i> | August 2016 |

Societies/Affiliations

- Dark Energy Spectroscopic Instrument (DESI) member (June 2023 - Present)
- LIGO Scientific Collaboration member (June 2015 - June 2023)
- American Physical Society

- Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)
- National Society of Hispanic Physicists

Student Mentoring

- **Kaitlyn Pak** Carnegie Mellon University
Undergraduate student - Multi-messenger associations with GWs
September 2025 - Present
- **Ariel Amsellem** Carnegie Mellon University
PhD student - Standard siren cosmology.
September 2023 - Present
- **Simon Yeung** University of Wisconsin-Milwaukee
PhD student - GW population systematics.
September 2023 - Present
- **Julia Yoon** Carnegie Mellon University
Undergraduate student - Dark siren cosmology with GW170817
September 2024 - May 2025
- **Diego Michel** Carnegie Mellon University
Undergraduate student - Gaussian processes for GW populations.
September 2023 - May 2024
- **Anarya Ray** University of Wisconsin-Milwaukee
PhD student (co-advised) - Binned Gaussian processes for GW populations.
September 2021 - June 2024
- **Pratyusava Baral** University of Wisconsin-Milwaukee
PhD student (co-advised) - Localization of BNS events with 3G GW detectors.
September 2021 - May 2023
- **Brandon Piotrkowski** University of Wisconsin-Milwaukee
PhD student (co-advised) - Multi-messenger association statistics.
September 2019 - February 2020

Skills

- **Programming skills:**
 - Proficiency with: Python (NumPy, SciPy, Pandas, Scikit-learn), JAX, C, Mathematica, LaTeX.
 - Familiar with: Stan, C++, MATLAB/Simulink.
 - LVK related software: Bilby, LALSuite, PyCBC, ligo.skymap, gwcosmo, hanabi.
- **Data Analysis:**
 - Bayesian model selection and parameter estimation, Monte Carlo methods.
 - Digital signal processing, machine learning, and basic numerical relativity experience.
- **Languages:** English (fluent), Spanish (fluent), and Korean (intermediate).

References

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